# Theoretical Alignment Document

Prototype: Understanding Arrays: Interactive Learning Module (Figma Prototype)

Target Topic: Basic data structures

Learner Audience: 1st-year IT/CS students with no prior programming background, but familiar with everyday sequential lists for example grocery lists, seat numbers.

SMART Objective: By the end of the module, learners will correctly identify and access array elements using index notation with 90% accuracy on a 5-item practice quiz, within a 15-minute session.

## Theorist Mapping

The wireframe's four-screen flow was deliberately structured so each screen operationalizes a specific Week 7 learning principle, rather than treating theory as decoration layered on top of the UI.

## 1. Edward Thorndike Law of Readiness

Core Idea: Learning is effective when the learner is mentally prepared and the new material connects to existing knowledge.

Where it appears: Screen 1 (Warm-Up) The mailbox/parking-slot analogy acts as a "meaning bridge," letting learners map a familiar real-world index system onto the abstract concept of arrays before any code is shown.

## 2. John Dewey Learning by Doing

Core Idea: Knowledge is constructed through direct, hands-on experience and multi-sensory interaction, not passive reading.

Where it appears: Screen 2 (Drag-and-Drop) Learners physically drag value blocks into indexed slots and get real-time visual and audio feedback, turning the abstract concept of indexing into an experienced action rather than a definition read on screen.

## 3. Thorndike Law of Exercise

Core Idea: Connections between stimulus and response are strengthened through repeated, guided practice.

Where it appears: Screen 3 (Guided Practice) The array-indexing exercise "Which box does array[2] point to?" gives learners a low-stakes, repeatable task with a hint option, reinforcing the index-to-value connection through active recall.

## 4. Thorndike Law of Effect / Psychological Safety

Core Idea: Responses followed by satisfaction are strengthened; responses followed by discomfort are weakened. Learning persists best in an emotionally safe environment.

Where it appears: Screen 4 (Feedback States) Success is met with a warm, celebratory modal; errors are met with a non-punitive, supportive modal like soft colors, no red/X icon, "Close! Counting starts at 0 want to try again?" that reframes mistakes as part of practice, encouraging retry rather than discouragement.

## 5. Herman Horne Facilitated Self-Activity

Core Idea: The teacher/system's role is to design an environment that facilitates the learner's own self-directed activity, not to transmit information directly.

Where it appears: Across Screens 2 to 4 The interface never simply states the rule for indexing; it structures an environment such as the draggable blocks, hint button, retry loop, progress stepper in which the learner discovers and self-corrects the rule through their own actions, with the system facilitating rather than lecturing.

## Design Rationale Summary

Read top to bottom, the four screens form a single instructional arc grounded in Week 7 theory: Readiness (Screen 1) opens the emotional and cognitive door, Active Learning (Screen 2) walks the learner through it via direct manipulation, Exercise (Screen 3) reinforces the pathway through repeatable practice, and the Law of Effect (Screen 4) determines whether that pathway is strengthened (through positive reinforcement) or safely revisited through empathetic error recovery with Horne's facilitated self-activity governing the interaction model throughout, since the system guides discovery rather than dictating answers.